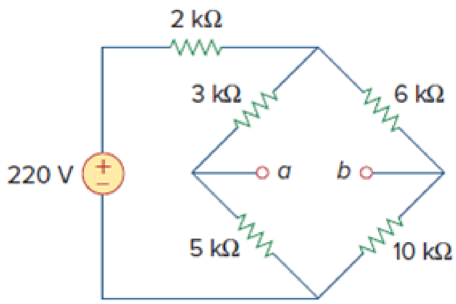
Chapter 4, Problem 92P

Problem

Consider the bridge circuit of Fig. 4.148. Is the bridge balanced? If the 10-kΩ resistor is replaced by an 18-kΩ resistor, what resistor connected between terminals *a*-*b* absorbs the maximum power? What is this power?



Step-by-step solution

Step 1 of 12

Refer to Figure 4.148 in the textbook.

Consider the ratio for a balanced bridge.

$$\frac{P}{Q}=\frac{R}{S} \quad \text{...... (1)}$$

Here,

P, *Q*, *R*, and *S*, are the bridge resistances.

Substitute $3\text{ k}\Omega$ for *P*, $6\text{ k}\Omega$ for *Q*, $5\text{ k}\Omega$ for *R*, and $10\text{ k}\Omega$ for *S*.

$$\frac{3}{6}=\frac{5}{10}$$
$$\frac{1}{2}=\frac{1}{2}$$

Hence, the bridge is $\boxed{\text{balanced}}$.

Step 2 of 12

Replace $10\text{ k}\Omega$ resistor by $18\text{ k}\Omega$.

Substitute $18\text{ k}\Omega$ for *S* in equation (1).

$$\frac{P}{Q}=\frac{R}{S}$$
$$\frac{3}{6}\neq\frac{5}{18}$$
$$\frac{1}{2}\neq\frac{5}{18}$$

Hence, the bridge is $\boxed{\text{not balanced}}$ on replacing the resistor of $10\text{ k}\Omega$ by $18\text{ k}\Omega$.

Step 3 of 12

Consider the following circuit diagram as shown in Figure 1.

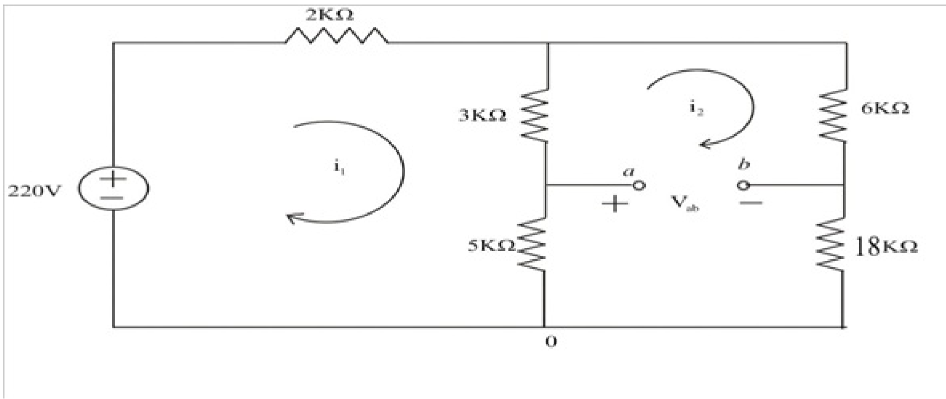


Figure 1

Step 4 of 12

In figure 1.

Here,

i_1, and i_2 are assumed to be in mA.

Apply KVL in loop 1.

$$220 = 2i_1 + 8(i_1 - i_2)$$

$$220 = 10i_1 - 8i_2 \quad \ldots\ldots (2)$$

Apply KVL in loop 2.

$$(5+3)(i_2 - i_1) + (6+18)(i_2) = 0$$
$$8(i_2 - i_1) + 24(i_2) = 0$$
$$32i_2 - 8i_1 = 0$$

$$i_1 = 4i_2 \quad \ldots\ldots (3)$$

Step 5 of 12

Substitute $4i_2$ for i_1 in equation (2).

$$220 = 10i_1 - 8i_2$$
$$220 = 10(4i_2) - 8i_2$$
$$220 = 32i_2$$
$$i_2 = 6.87 \text{ mA}$$

Substitute 6.87 mA for i_2 in equation (3).

$$\begin{aligned} i_1 &= 4i_2 \\ &= 4(6.87) \\ &= 27.48 \text{ mA} \end{aligned}$$

Step 6 of 12

Consider the following circuit, when shorting a, and b.

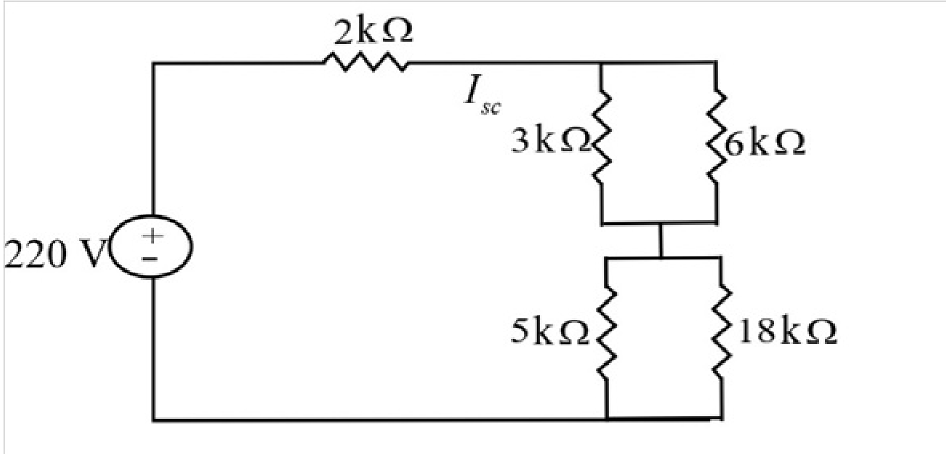


Figure 2

Step 7 of 12

Since, the resistance, $3\,\text{k}\Omega \| 6\,\text{k}\Omega$ and $5\,\text{k}\Omega \| 18\,\text{k}\Omega$, are in series.

The equivalent resistance *Z*, is given by,

$$
\begin{aligned}
Z &= \left(3\,\text{k}\Omega \| 6\,\text{k}\Omega\right) + \left(5\,\text{k}\Omega \| 18\,\text{k}\Omega\right) \\
&= \frac{(3)(6)}{3+6} + \frac{(5)(18)}{5+18} \\
&= 2 + 3.91 \\
&= 5.91\,\text{k}\Omega
\end{aligned}
$$

Step 8 of 12

The following is the modified circuit diagram of Figure 2 as shown in Figure 3.

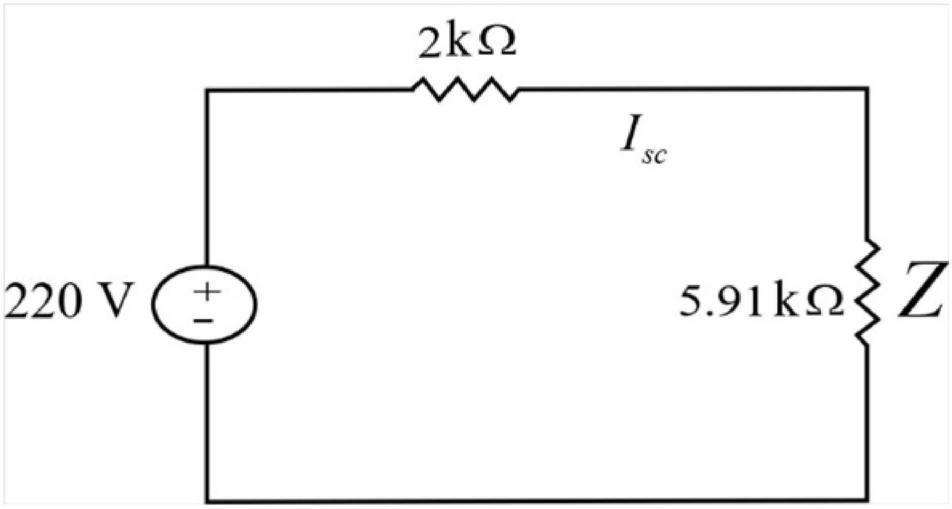


Figure 3

Step 9 of 12

Determine the short circuit current I_{sc}, that would flow when point a, and b are shorted as shown in Figure 3.

Apply KVL in Figure 3.

$$(2+5.91)I_{sc} = 220$$
$$I_{sc} = \frac{220}{7.91}$$
$$I_{sc} = 27.81 \text{ mA}$$

$$I_{sc} = 27.81 \text{ mA}$$

Step 10 of 12

Determine the open circuit voltage v_{oc}, across point a, and b using Figure 1.

$$v_{ab} = 5(i_1 - i_2) + 18(-i_2)$$
$$v_{ab} = 5i_1 - 23i_2$$

Here,

v_{ab} is same as v_{oc}, $v_{ab} = v_{oc}$.

Substitute 27.48 mA for i_1, and 6.87 mA for i_2.

$$\begin{aligned} v_{ab} &= 5(27.48) - 23(6.87) \\ &= 137.48 - 158.01 \\ &= -20.53 \text{ V} \end{aligned}$$

$$|v_{ab}| = 20.53 \text{ V}$$

Step 11 of 12

Determine the resistor connected between terminal $a-b$, R_{Th}.

$$\begin{aligned} R_{\text{Th}} &= \frac{v_{oc}}{I_{sc}} \\ &= \frac{v_{ab}}{I_{sc}} \end{aligned}$$

Here,

R_{Th} is the Thevenin resistance.

Substitute 20.53 V for $|v_{ab}|$, and 27.81 mA for I_{sc}.

$$\begin{aligned} R_{\text{Th}} &= \frac{20.53}{27.81 \times 10^{-3}} \\ &= 0.738 \text{ k}\Omega \end{aligned}$$

$$R_{\text{Th}} = 0.738 \text{ k}\Omega$$

Therefore, the resistor connected between terminal $a-b$, R_{Th} is $\boxed{0.738 \text{ k}\Omega}$.

Step 12 of 12

Determine the maximum absorb power by the resistor R_{Th}.

$$P_{max} = \frac{v_{ab}^2}{4R_{Th}}$$

Substitute $0.738\ \text{k}\Omega$ for R_{Th}, and $20.53\ \text{V}$ for $|v_{ab}|$.

$$\begin{aligned} P_{max} &= \frac{(20.53)^2}{4(0.738 \times 10^3)} \\ &= \frac{421.48}{2952} \\ &= 0.143\ \text{W} \end{aligned}$$

$$P_{max} = 0.143\ \text{W}$$

Therefore, the maximum absorb power by the resistor R_{Th} is $\boxed{P_{max} = 0.143\ \text{W}}$.